

Operational Research II, Final Exam, 2026 Spring

- Duration: 120 minutes
- Closed material, No calculator

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- Write legibly.
- Justification is necessary unless stated otherwise.

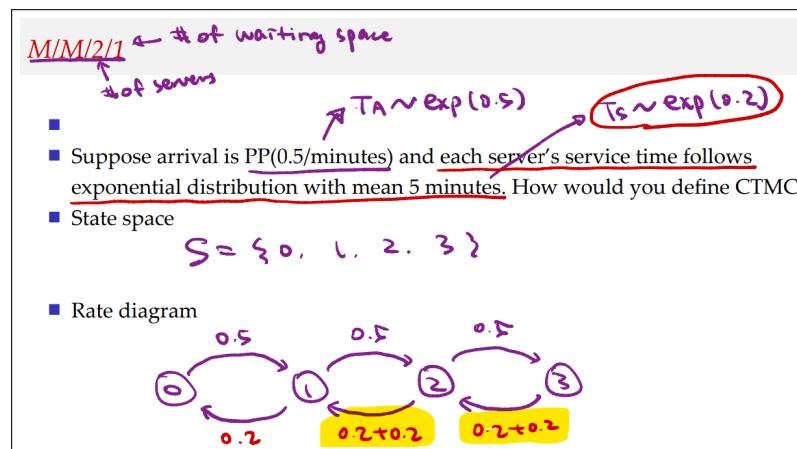
1	20
2	20
3	30
Total	70

#1. Assume that calls arrive at a customer call center between 9am ($t = 0$) and 2pm ($t = 5$) following a non-homogeneous Poisson process with the rate function of the following:

$$\lambda(t) = \begin{cases} 4 & 0 \leq t < 1 \\ 2t + 2 & 1 \leq t < 3 \\ 8 & 3 \leq t < 5 \end{cases}$$

- (a) What is the probability that the call center receives 7th call before 10am ($t = 1$)? [10pts]
- (b) What is the expected number of arrivals between 11am ($t = 2$) and 1pm ($t = 4$)? [10pts]

#2. Consider the following lecture note content. To define the $M/M/2/1$ queue as a CTMC, the state is defined as the number of customers in the system, yielding the state space $S = \{0, 1, 2, 3\}$. A corresponding rate diagram has also been constructed.



(a) In the rate diagram, the highlighted portions are labeled as “ $0.2 + 0.2$ ”. This indicates that when there are 2 or 3 customers in the system, both servers are busy, and thus the rate at which the number of customers decreases is the sum of the service rates of the two servers. Formally state the relevant theorem that justifies this modeling approach. (Hint: The theorem is related to the exponential distribution.) [10pts]

(b) Prove the theorem you stated above. [10pts]

#3. Consider a very small McDonald's drive-through. An arriving vehicle must go through two stations: an **order** station, where the vehicle places an order and makes payment, and a **pickup** station, where the vehicle receives its food.

The movement of vehicles is described as follows.

- If the **order** station is empty upon arrival, the vehicle immediately enters the **order** station and begins service.
- If an arriving vehicle finds the **order** station occupied, it leaves the facility immediately. In other words, there is no waiting space.
- When a vehicle completes service at the **order** station and finds the **pickup** station empty, it proceeds to the **pickup** station.
- When a vehicle completes service at the **order** station and finds the **pickup** station occupied, the vehicle remains at the **order** station and waits until the **pickup** station becomes empty. In this case, the **order** station is still occupied, so no newly arriving vehicle cannot enter the **order** station.
- When a vehicle reaches the **pickup** station, it receives its food and then immediately leaves the facility as soon as completing service.

The following assumptions are made for this queueing system.

- Each station can serve at most one vehicle at a time.
- Vehicles arrive according to a Poisson process with rate 1 vehicle per minute.
- The **order** station can process, on average, 1 vehicle per minute, and the **pickup** station can also process, on average, 1 vehicle per minute. The service times at both stations are exponentially distributed.
- All interarrival and service times are independent.

(a) Define and describe the states of the CTMC. Then, draw the transition diagram¹. [10pts]

(b) In the long run, what percentage of arriving customer vehicles receive the service? [10pts]

(c) The owner is considering adding one waiting space before the **order** station. If the same type of CTMC analysis is to be performed for this modified system, define the additional states that must be introduced. [10pts]

¹Hint: Your state definition should distinguish between the following two cases: i) a vehicle at the order station is currently placing an order; ii) a vehicle at the order station has already finished ordering but is blocked because the pickup station window is occupied.

[다음은 3번 문제의 한글 번역이다]

매우 작은 맥도날드 드라이브스루를 고려하자. 도착한 차량은 두 개의 station을 순서대로 거쳐야 한다. 첫 번째는 차량이 주문과 결제를 수행하는 **order station**이고, 두 번째는 차량이 음식을 수령하는 **pickup station**이다. 차량의 이동 방식은 다음과 같다.

1. 도착한 차량이 **order station**이 비어 있는 것을 발견하면, 그 차량은 즉시 **order station**으로 진입하여 서비스를 시작한다.
2. 도착한 차량이 **order station**이 점유되어 있는 것을 발견하면, 그 차량은 즉시 facility를 떠난다. 즉, waiting space는 존재하지 않는다.
3. **order station**에서 서비스를 완료한 차량이 **pickup station**이 비어 있는 것을 발견하면, 그 차량은 **pickup station**으로 이동한다.
4. **order station**에서 서비스를 완료한 차량이 **pickup station**이 점유되어 있는 것을 발견하면, 그 차량은 **pickup station**이 빌 때까지 **order station**에 남아 기다린다. 이 경우 **order station**은 여전히 점유된 상태이므로, 새로 도착한 차량은 **order station**에 진입할 수 없다.
5. 차량이 **pickup station**에 도달하면 음식을 수령하고, 서비스를 완료하는 즉시 facility를 떠난다.

이 대기행렬 시스템에 대해 다음과 같은 가정을 둔다.

- 각 station은 한 번에 최대 한 대의 차량만 서비스할 수 있다.
- 차량은 분당 1대의 비율을 갖는 Poisson process에 따라 도착한다.
- **order station**은 평균적으로 분당 1대의 차량을 처리할 수 있으며, **pickup station**도 평균적으로 분당 1대의 차량을 처리할 수 있다. 두 station에서의 서비스 시간은 모두 exponential distribution을 따른다.
- 모든 interarrival time과 service time은 서로 독립이다.

(a) CTMC의 state를 정의하고 설명하라. 그런 다음 transition diagram을 그려라². [10pts]

(b) 장기적으로, 도착한 customer 차량 중 몇 퍼센트가 full service를 받는가? [10pts]

(c) 점주는 **order station** 앞에 waiting space를 하나 추가하는 방안을 고려하고 있다. 이 수정된 시스템에 대해 동일한 방식으로 CTMC 분석을 수행하고자 할 때, 새롭게 추가되어야 하는 state들을 정의하라. [10pts]

²힌트: State를 정의할 때 다음 두 경우를 구분해야 한다. i) **order station**에 있는 차량이 현재 주문을 수행하고 있는 경우; ii) **order station**에 있는 차량이 이미 주문을 완료했지만, **pickup station**이 점유되어 있어 blocked 상태에 있는 경우.